

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("train (1).csv")

df.head()
df.shape
df.info()
df.describe()
df.isnull().sum()
df.nunique()

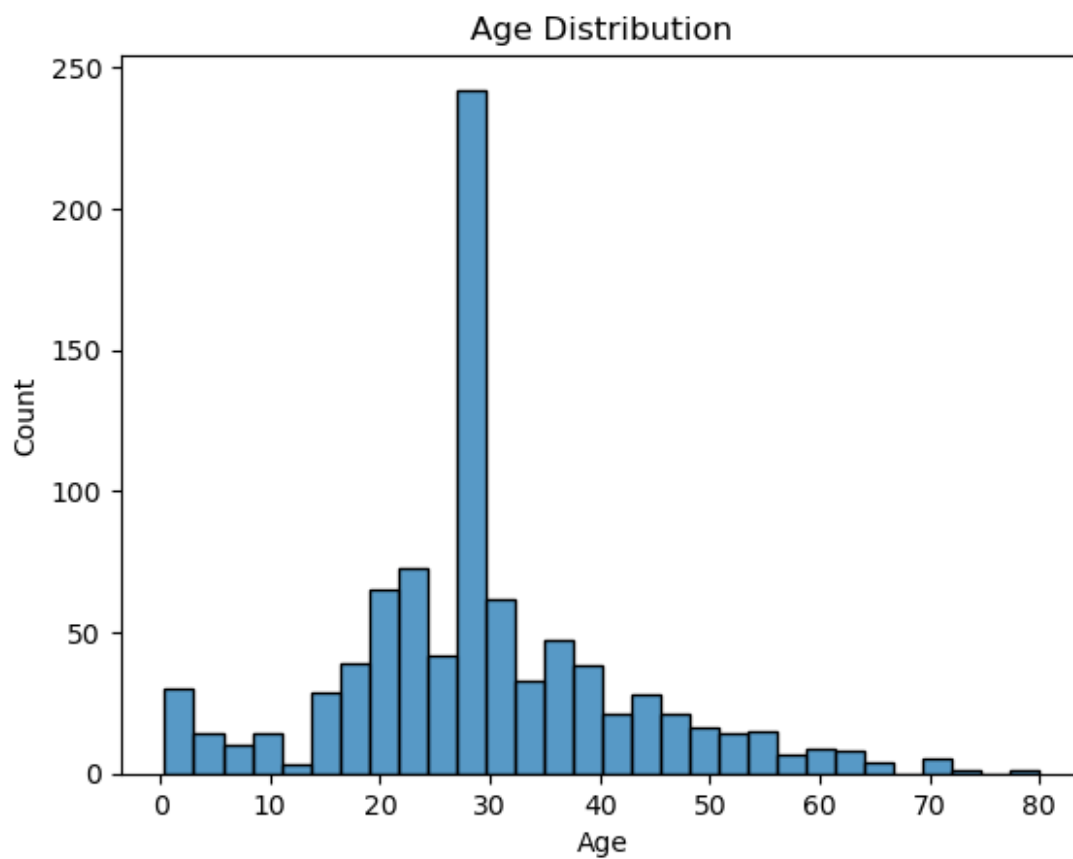
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   PassengerId           891 non-null   int64
1   Survived              891 non-null   int64
2   Pclass               891 non-null   int64
3   Name                 891 non-null   object
4   Sex                 891 non-null   object
5   Age                714 non-null   float64
6   SibSp              891 non-null   int64
7   Parch             891 non-null   int64
8   Ticket            891 non-null   object
9   Fare             891 non-null   float64
10  Cabin            204 non-null   object
11  Embarked        889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

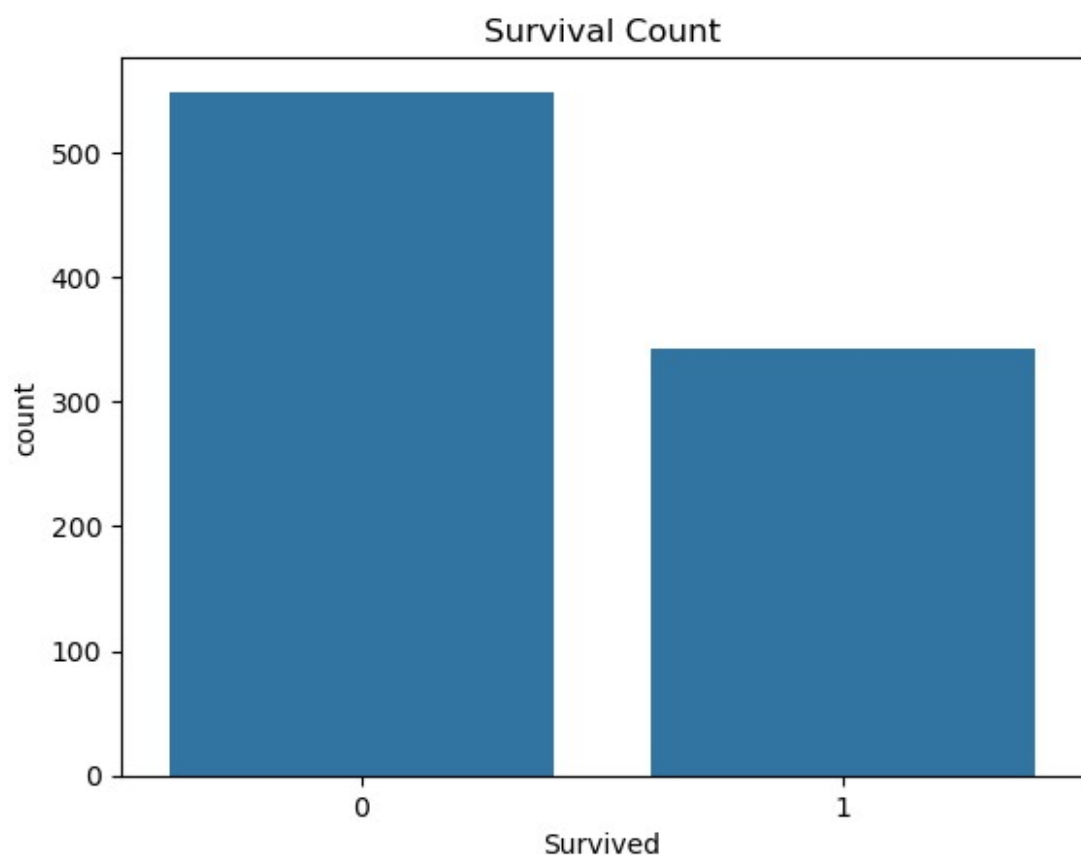
PassengerId    891
Survived        2
Pclass          3
Name           891
Sex             2
Age            88
SibSp           7
Parch           7
Ticket         681
Fare           248
Cabin          147
Embarked        3
dtype: int64

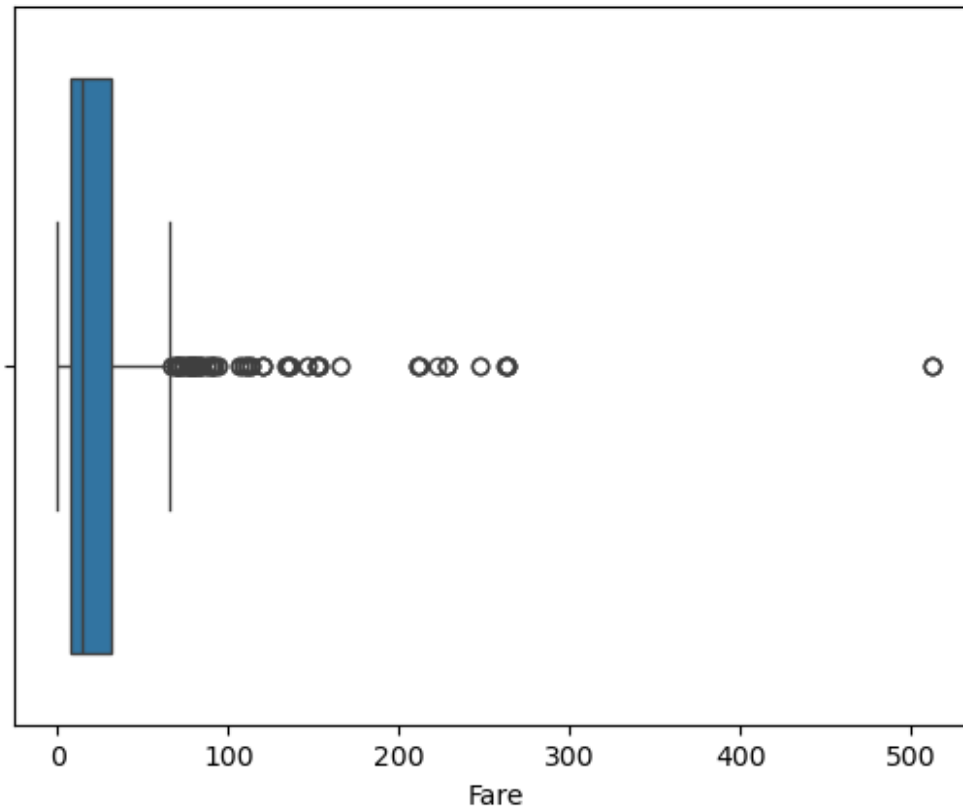
# Fix chained assignment warnings
df['Age'] = df['Age'].fillna(df['Age'].median())

```

```
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])  
  
# drop Cabin if it exists  
if 'Cabin' in df.columns:  
    df.drop(columns=['Cabin'], inplace=True)  
  
sns.histplot(df['Age'], bins=30)  
plt.title("Age Distribution")  
plt.show()  
  
sns.countplot(x='Survived', data=df)  
plt.title("Survival Count")  
plt.show()  
  
sns.boxplot(x=df['Fare'])  
plt.show()
```



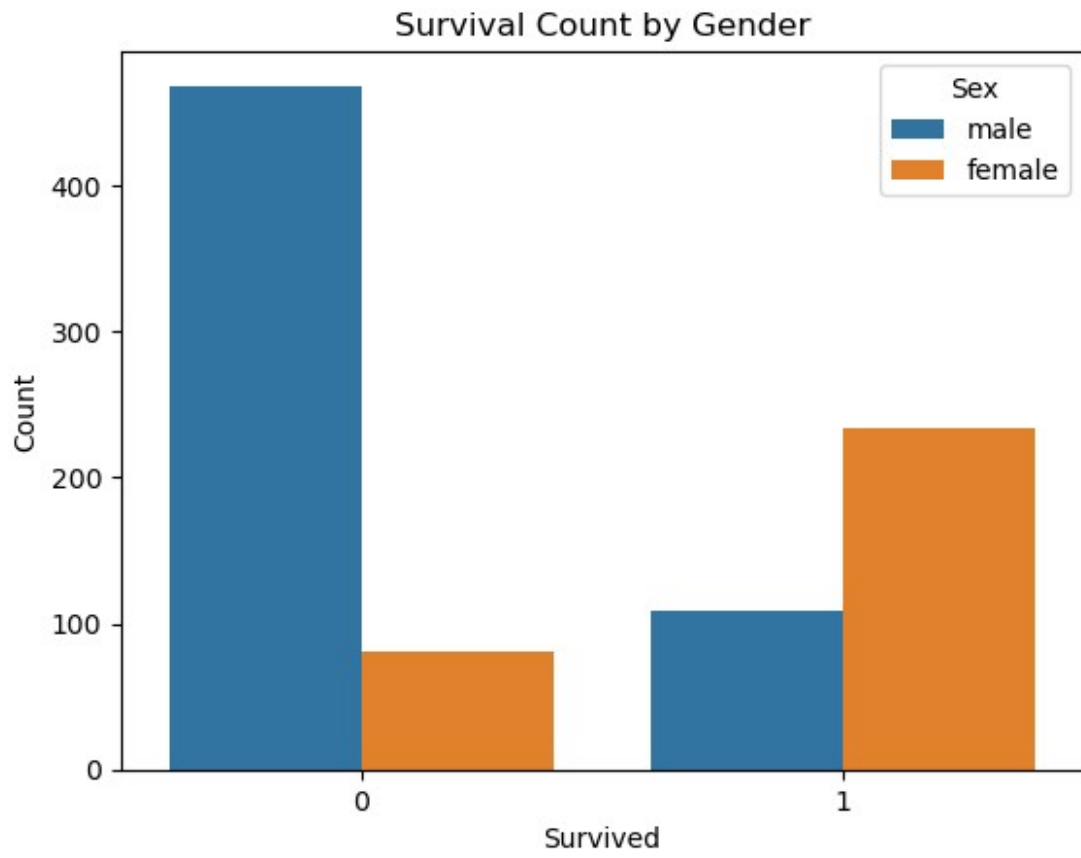


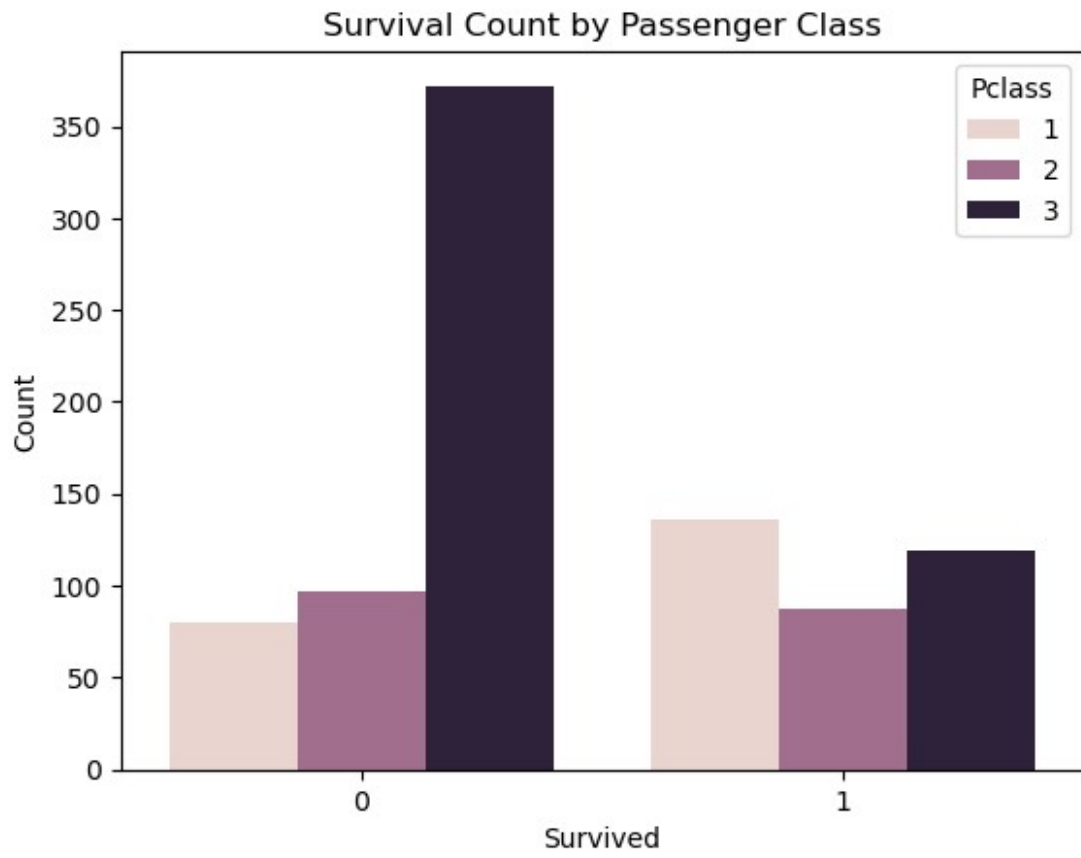


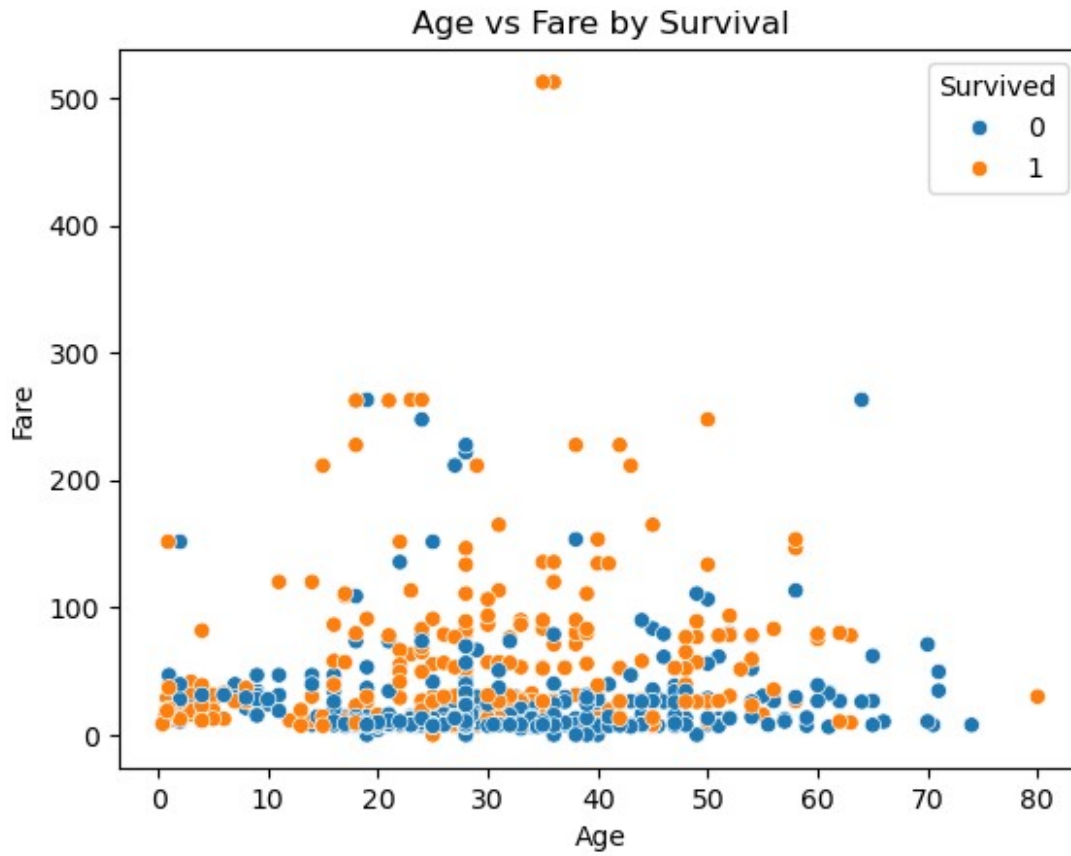
```
# Survival by Sex
sns.countplot(x='Survived', hue='Sex', data=df)
plt.title("Survival Count by Gender")
plt.xlabel("Survived")
plt.ylabel("Count")
plt.show()

# Survival by Pclass
sns.countplot(x='Survived', hue='Pclass', data=df)
plt.title("Survival Count by Passenger Class")
plt.xlabel("Survived")
plt.ylabel("Count")
plt.show()

# Age vs Fare
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
plt.title("Age vs Fare by Survival")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```

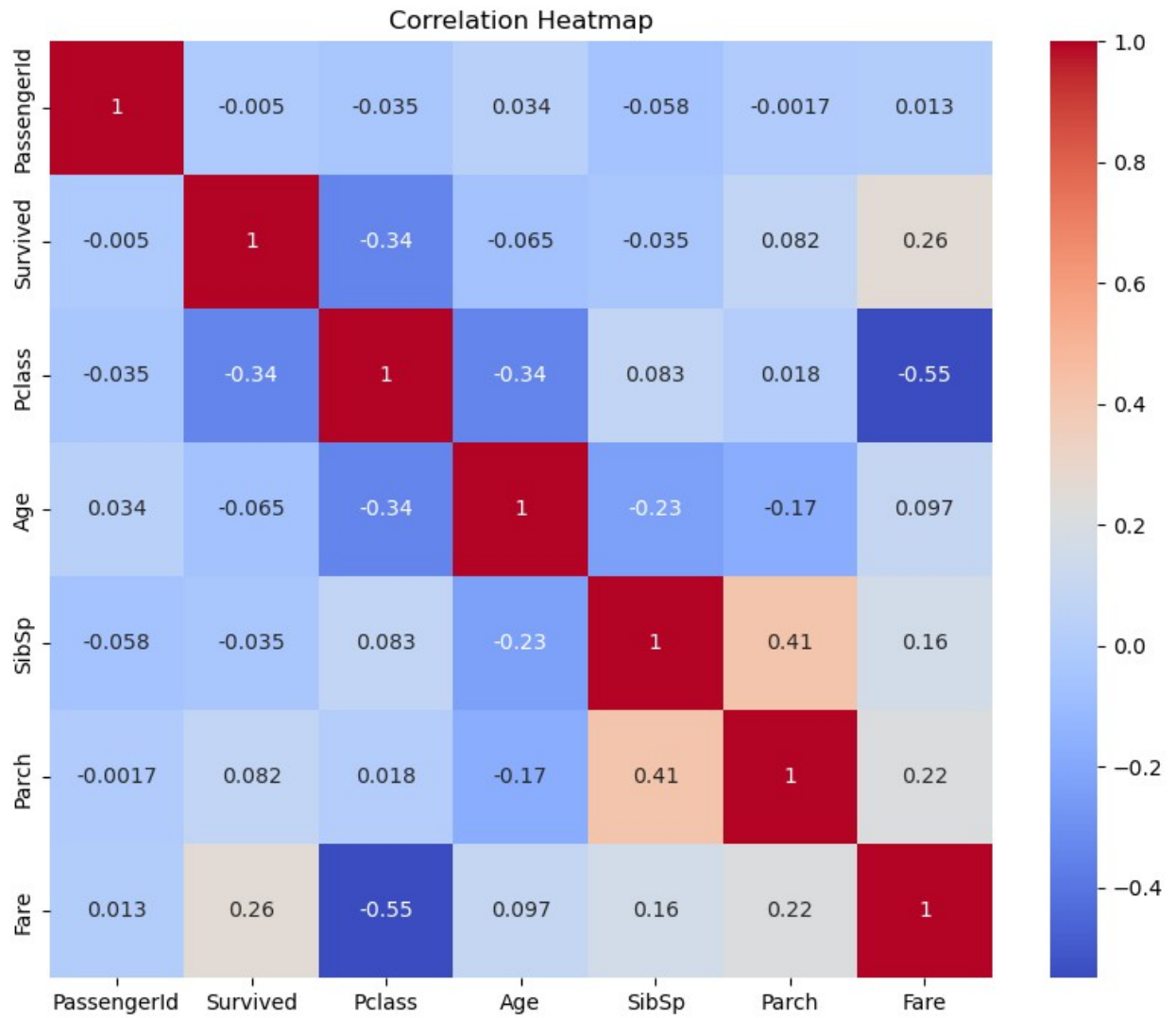






```
# Heatmap of correlations
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()

# Pairplot of selected features
sns.pairplot(df[['Age', 'Fare', 'Pclass', 'Survived']],
             hue='Survived')
plt.suptitle("Pairplot: Age, Fare, Pclass vs Survived", y=1.02)
plt.show()
```



Pairplot: Age, Fare, Pclass vs Survived

